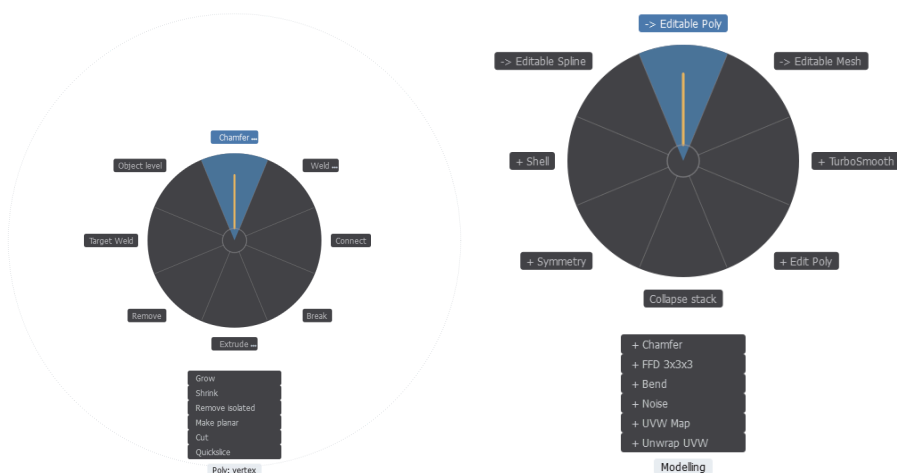


# MarkingForge

Marking menus, hotbox and gesture sliders for Autodesk 3ds Max 2027

INSTALLATION AND CONFIGURATION GUIDE · VERSION 1.0.0



Hold a key, flick the mouse, done. This guide covers installing the plugin, setting up shortcuts, using the 24 built-in dials, changing them, and adding MarkingForge commands to 3ds Max's menus, toolbars and quad menus. Every option in detail, with tips and troubleshooting, is in the companion document **MarkingForge Reference**.

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# 1. What you get

The download is one ZIP file. Unpacked, it holds:

|                                      |                                             |
|--------------------------------------|---------------------------------------------|
| MarkingForge_1.0.0\                  |                                             |
| MarkingForge_1.0.0.mzp               | installer - drag onto a viewport (method 1) |
| Documentation\                       |                                             |
| MarkingForge_Installation_Guide.pdf  | this document                               |
| MarkingForge_QuickStart.pdf          | tutorials: first gestures, the editor       |
| MarkingForge_Reference.pdf           | every option, tips, troubleshooting         |
| MarkingForge_Shortcut_Card.pdf       | the 24 dials on one page to print           |
| MarkingForge_for_Maya_Users.pdf      | Maya habits mapped to MarkingForge          |
| MarkingForge_Preset_Packs.pdf        | the four ready-made dial sets               |
| MarkingForge_Studio_Deployment.pdf   | silent install for studios                  |
| Presets\*.json                       | Modelling, Animation, Look-dev, Archviz     |
| Studio\Deploy-MarkingForge.ps1       | silent install / update / removal           |
| README.txt                           | the short version of this document          |
| CHANGELOG.txt                        | what is in this version                     |
| LICENSE.txt                          | licence agreement                           |
| Manual install\                      |                                             |
| MarkingForge_Install.ms              | the same installer as a script (method 2)   |
| MarkingForge\                        | the plugin folder itself (method 3)         |
| PackageContents.xml                  | tells 3ds Max what to load                  |
| plugins\MarkingForge.gup             | the plugin (C++)                            |
| scripts\MarkingForge_Startup.ms      | builds the MarkingForge menu                |
| macroscripts\MarkingForge_Macros.mcr | the menu commands                           |
| tools\mf_editor\*.py                 | the menu editor (Python)                    |
| Documentation\*.pdf                  | these guides - MarkingForge > Documentation |

## What kind of plugin is it?

MarkingForge is a **native C++ plugin** - a Global Utility Plugin, the `.gup` file, which is a Windows DLL that 3ds Max loads when it starts. The dials, the gesture recognition and the drawing all run in C++ with Qt, the same toolkit 3ds Max itself is built with, so a dial appears in a few milliseconds.

Around it sit three small parts: a **startup script** (MAXScript) that adds the MarkingForge menu to the menu bar, a set of **macroscripts** (MAXScript) behind the menu commands - which is also what lets you put them on toolbars and quad menus - and the **menu editor**, written in Python with PySide6, both of which ship with 3ds Max 2027. Nothing else needs installing.

Everything is installed as one folder that 3ds Max loads through its standard *Application Plugins* mechanism. Nothing is copied into the 3ds Max program folder, and nothing is installed outside your user profile unless you choose "All users".

## 2. Requirements

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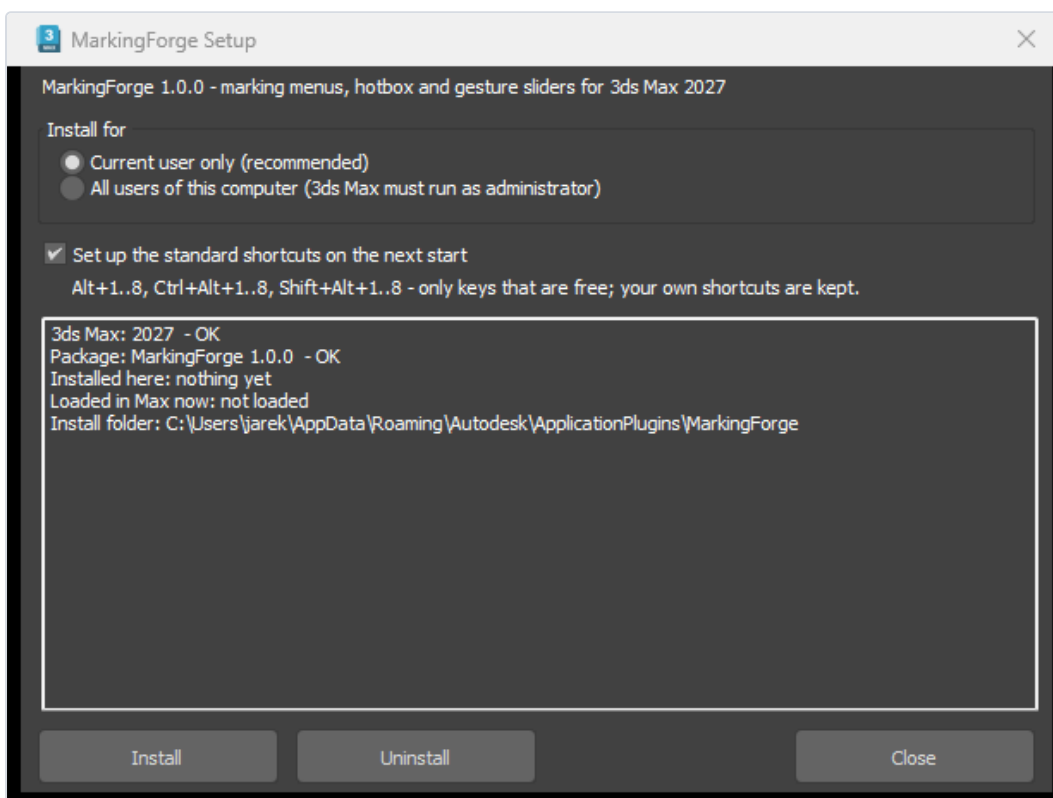
| Item               | Requirement                                                                                                                                                                                                    |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3ds Max            | Autodesk 3ds Max 2027, Windows 64-bit. This build does not load in other versions; the installer checks and refuses.                                                                                           |
| Windows            | Windows 10 or 11, 64-bit.                                                                                                                                                                                      |
| Visual C++ runtime | Microsoft Visual C++ 2015-2022 Redistributable (x64). 3ds Max installs it itself, so it is almost always present. If the plugin does not load, reinstall it from Microsoft ( <code>vc_redist.x64.exe</code> ). |
| Rights             | None special for "Current user". "All users" needs 3ds Max started as administrator.                                                                                                                           |

### 3. Installation - three methods

All three methods end in the same place: the folder `MarkingForge` in `%APPDATA%\Autodesk\ApplicationPlugins\`. Pick the one you like; methods 1 and 2 run the same installer.

#### Method 1 - drag and drop the .mzp RECOMMENDED

- 1 Unzip the download and start 3ds Max 2027.
- 2 Drag `MarkingForge_1.0.0.mzp` from Explorer and drop it onto any viewport.
- 3 The **MarkingForge Setup** window opens. Leave "Current user only" selected and press **Install**.
- 4 Restart 3ds Max.



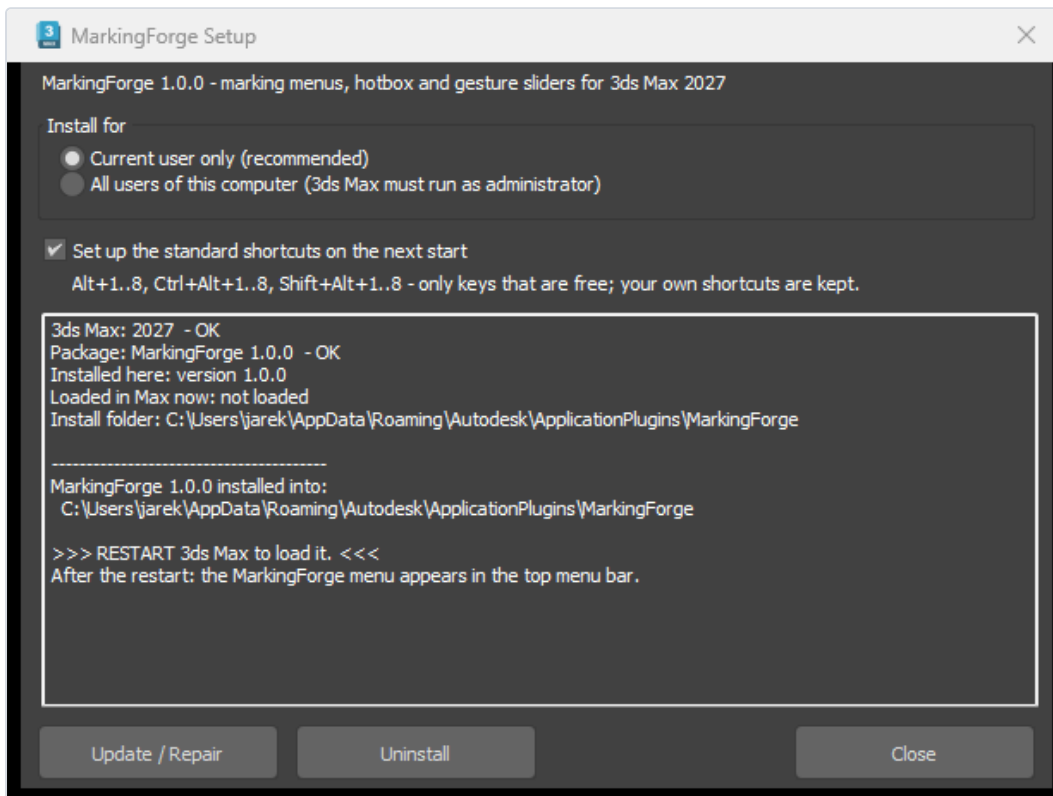
The MarkingForge Setup window. It checks the 3ds Max version, the package and any other copy of the plugin before it lets you install.

#### Method 2 - run the installer script

- 1 Unzip the download.
- 2 In 3ds Max choose **Scripting > Run Script...**
- 3 Open `Manual\install\MarkingForge_Install.ms`.
- 4 The same Setup window opens - press **Install**, then restart 3ds Max.

## Method 3 - copy the folder yourself

- 1 Close 3ds Max.
- 2 Open Explorer, paste `%APPDATA%\Autodesk\ApplicationPlugins` into the address bar and press Enter (create the `ApplicationPlugins` folder if it does not exist).
- 3 Copy the folder `Manual install\MarkingForge` into it.
- 4 Start 3ds Max.



After Install: the Setup window says where the plugin went and asks for a restart.

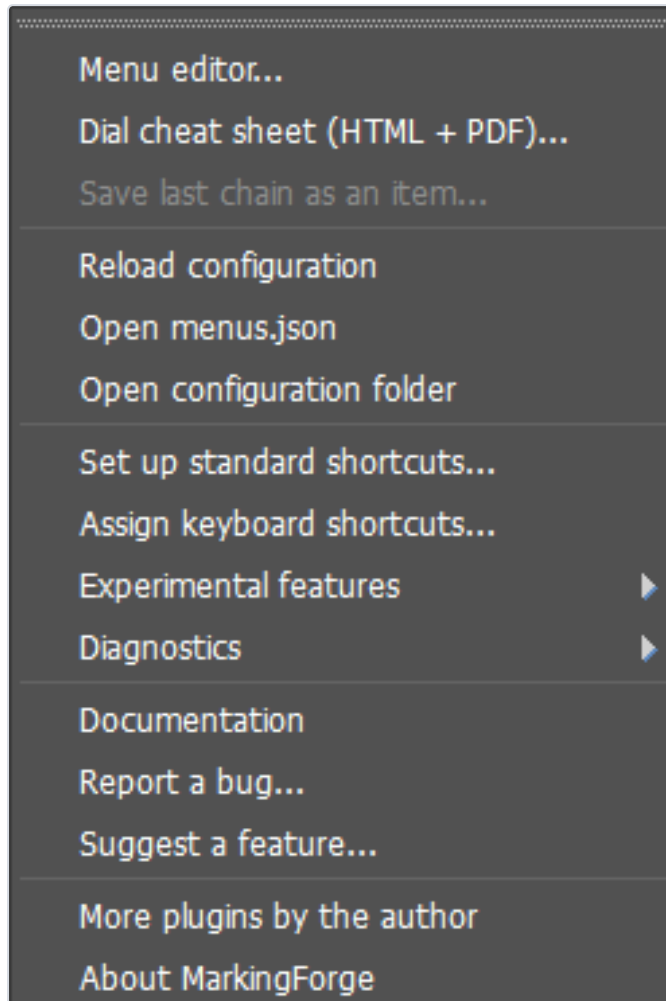
## Setup window options

| Option                                          | What it does                                                                                                                                                                                                                                |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Current user only                               | Installs into <code>%APPDATA%\Autodesk\ApplicationPlugins\MarkingForge</code> . No administrator rights needed.                                                                                                                             |
| All users of this computer                      | Installs into <code>%ProgramData%\Autodesk\ApplicationPlugins\MarkingForge</code> . Start 3ds Max as administrator first.                                                                                                                   |
| Set up the standard shortcuts on the next start | On the first start after installing, every dial gets its key - <code>Alt+1..8</code> , <code>Ctrl+Alt+1..8</code> , <code>Shift+Alt+1..8</code> . A key you already use for something else is skipped, and no shortcut of yours is changed. |
| Install / Update / Repair                       | Copies the plugin. Over an existing installation it replaces it, keeping your dials and settings. It works even while the old version is loaded.                                                                                            |
| Uninstall                                       | Removes the plugin folder. Your own settings are kept (see chapter 12).                                                                                                                                                                     |

**Why a restart?** MarkingForge is a C++ plugin, and 3ds Max loads C++ plugins only while it starts. The installer copies the files; the next start loads them.

## 4. Checking that it works

After the restart a **MarkingForge** menu appears in the menu bar, before *Help*.

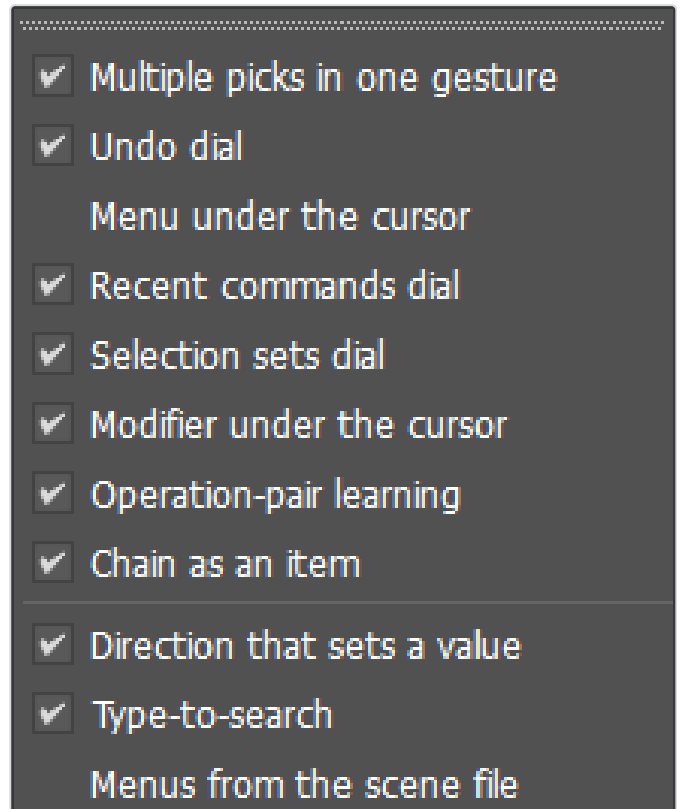


The MarkingForge menu.

To be certain, open the MAXScript Listener (F11) and type:

```
MarkingForge.version()
```

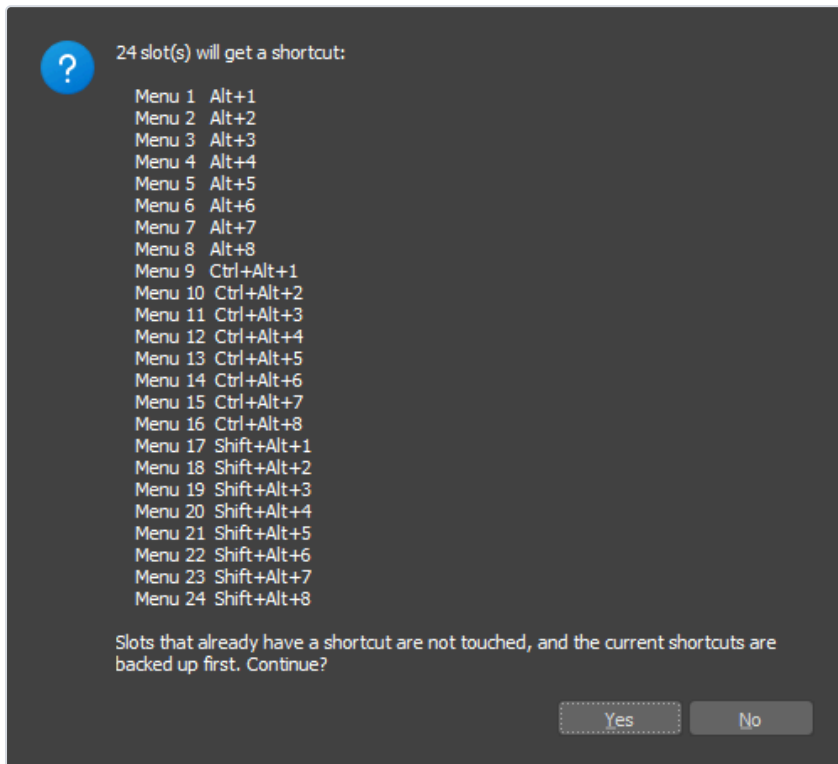
It prints the version number. "Unknown property" means the plugin was not loaded - see Troubleshooting.



Its Experimental features submenu - options you switch on deliberately.

## 5. Keyboard shortcuts

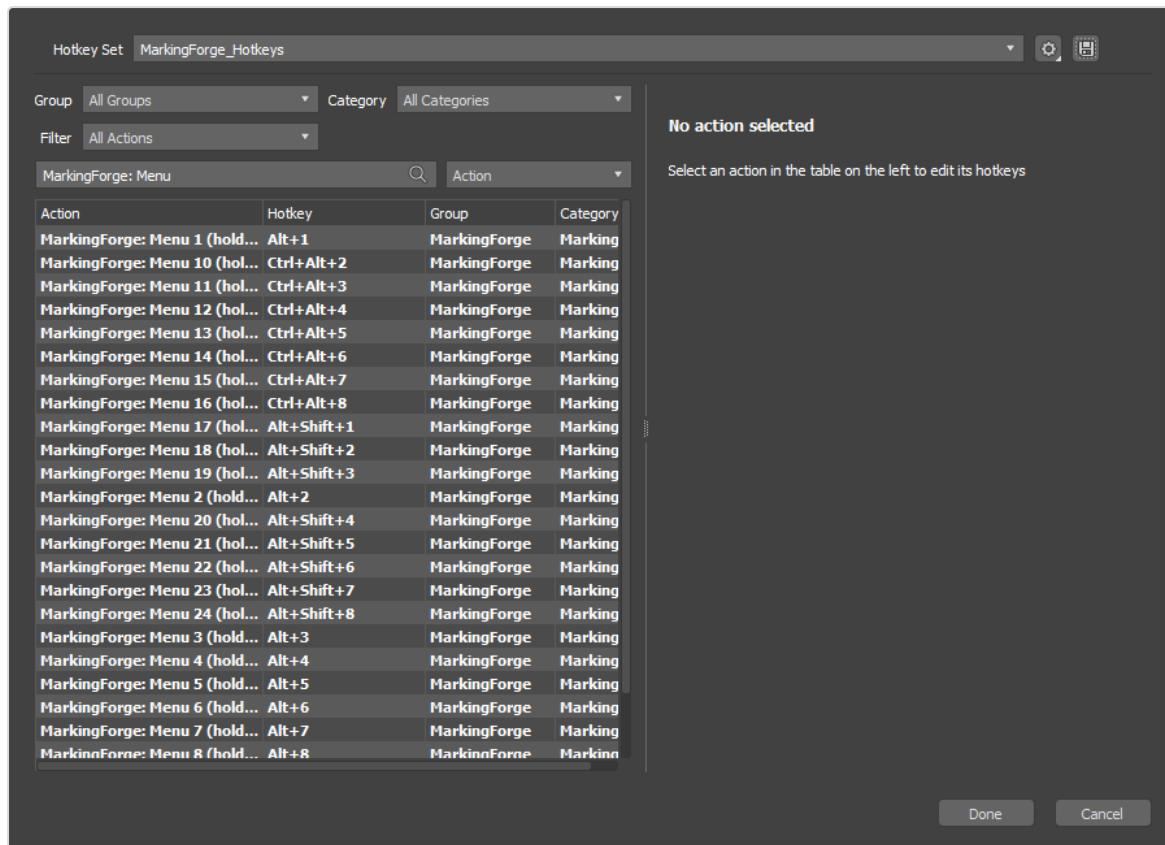
Each of the 24 dials opens by holding a key. If you kept the installer option on, they are already set. Otherwise - or to fill in dials added later - choose **MarkingForge > Set up standard shortcuts...** It shows exactly what it will do and asks once:



Set up standard shortcuts: only dials without a shortcut, only keys nothing else uses.

| Key          | Dial                | Key               | Dial             | Key                | Dial               |
|--------------|---------------------|-------------------|------------------|--------------------|--------------------|
| <b>Alt+1</b> | Modelling           | <b>Ctrl+Alt+1</b> | Transform        | <b>Shift+Alt+1</b> | Materials          |
| <b>Alt+2</b> | Hotbox - Max's menu | <b>Ctrl+Alt+2</b> | Modifiers        | <b>Shift+Alt+2</b> | Animation          |
| <b>Alt+3</b> | Create              | <b>Ctrl+Alt+3</b> | Show and hide    | <b>Shift+Alt+3</b> | Scene              |
| <b>Alt+4</b> | View                | <b>Ctrl+Alt+4</b> | Viewports        | <b>Shift+Alt+4</b> | File               |
| <b>Alt+5</b> | Selection           | <b>Ctrl+Alt+5</b> | Viewport display | <b>Shift+Alt+5</b> | Selection sets     |
| <b>Alt+6</b> | Modifier stack      | <b>Ctrl+Alt+6</b> | Snaps and grid   | <b>Shift+Alt+6</b> | UV mapping         |
| <b>Alt+7</b> | Undo                | <b>Ctrl+Alt+7</b> | Align and pivot  | <b>Shift+Alt+7</b> | Tools and setup    |
| <b>Alt+8</b> | Recent commands     | <b>Ctrl+Alt+8</b> | Render           | <b>Shift+Alt+8</b> | Lights and cameras |

You can change any of them in the menu editor ("This slot's shortcut") or in 3ds Max's own **Customize > Hotkey Editor**, where they are listed as *MarkingForge: Menu 1 ... Menu 24*:



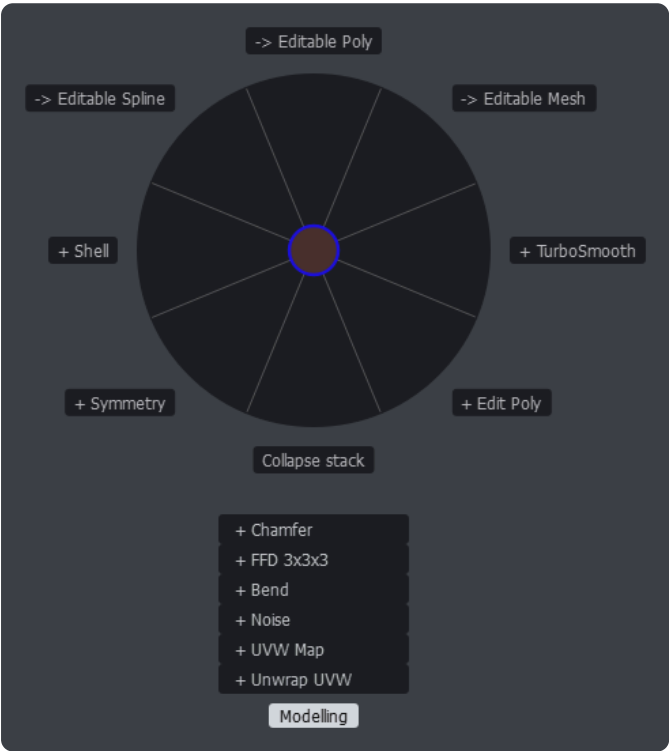
The Hotkey Editor filtered to MarkingForge. Any key can be reassigned here as well.

## 6. Using the dials

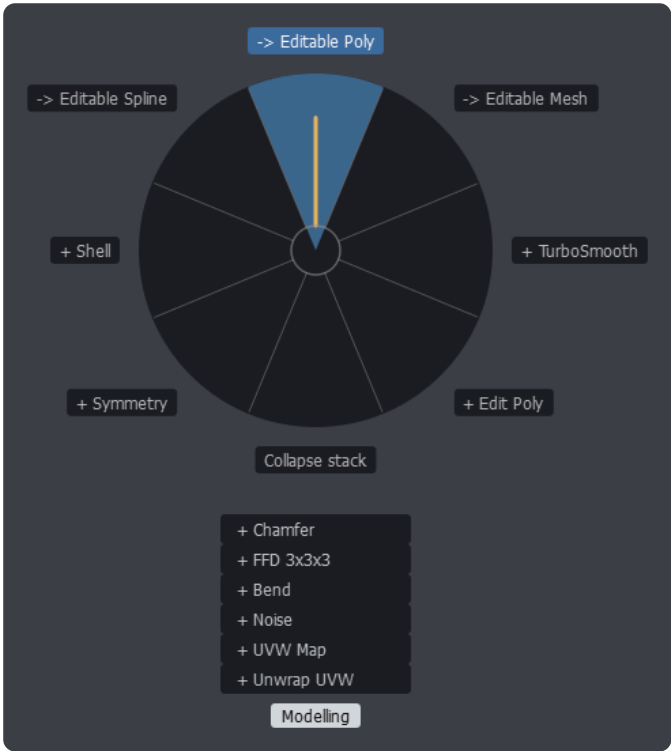
The same key does three things, depending on the hand:

| Move         | What happens                                                                                                                                                                 |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hold</b>  | Hold the key: the dial appears around the cursor. Move towards an item and release the key. For learning and browsing.                                                       |
| <b>Flick</b> | Press the key and move the mouse at once, without waiting. The item runs without the dial being drawn. After a week or two you stop looking - the direction is in your hand. |
| <b>Tap</b>   | Press and release without moving. Does nothing by default; the editor can give it an item.                                                                                   |

Releasing in the centre cancels. So do **Esc** and the right mouse button.



Alt+1 on an ordinary object: convert and add modifiers. Less common items sit in a list under the ring - rows are clicked, directions are aimed at.



Aiming north: the direction lights up. Releasing runs *Convert to Editable Poly*.

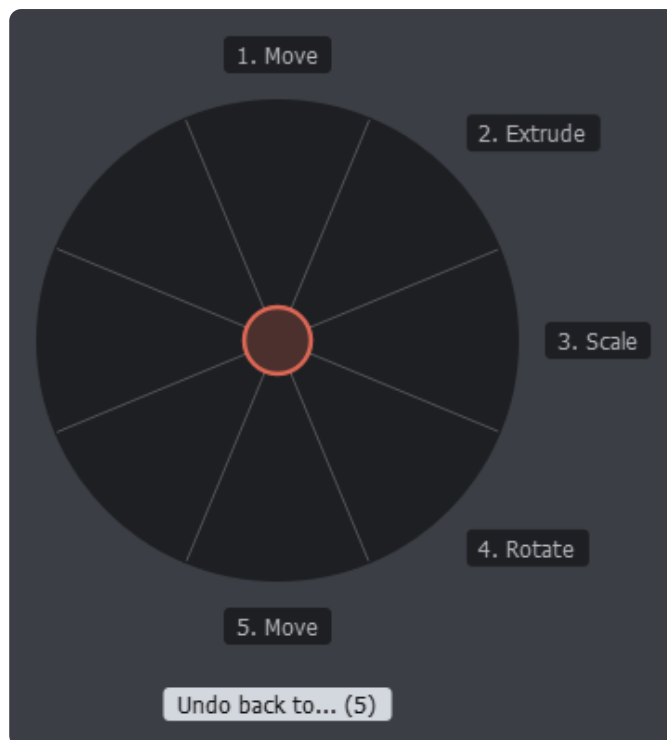
Commands with a settings dialog (the caddy) - Chamfer, Extrude, Inset... - open it on an ordinary release, like the settings icon in a quad menu. Moving **far past the ring** before releasing runs them **without** the dialog - "far" is the **dotted circle** outside the dial, which brightens once you are past it. Either can be made the default per command (**Settings...** in the editor).

More tricks - several commands in one gesture, the dial of the object under the cursor, the suggested next command and search by typing - are in the **QuickStart** ("Five tricks worth knowing") and in chapter 2 of the **Reference**.

**Alt+2** is the **hotbox**: all of 3ds Max's main menu as tiles around the cursor, including menus other plugins add - point at a menu title, then at an entry. **Alt+7** is the **undo dial**: the last eight operations by name; N undoes one step, NE two, and so on.



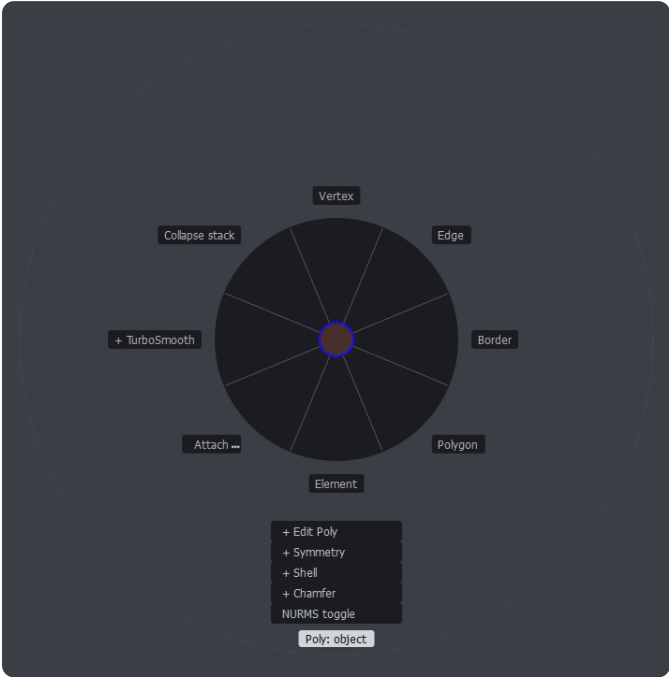
Alt+2 - the hotbox.



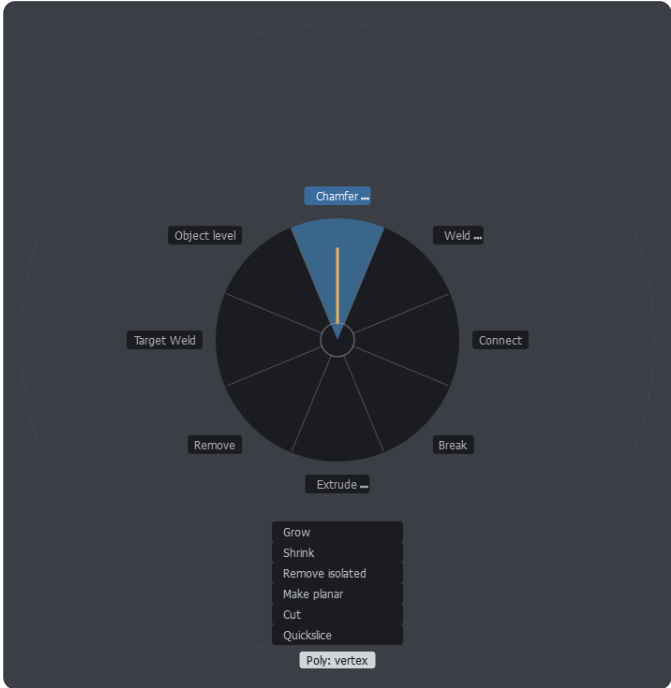
Alt+7 - the undo dial.

# 7. Dials that follow what you are doing

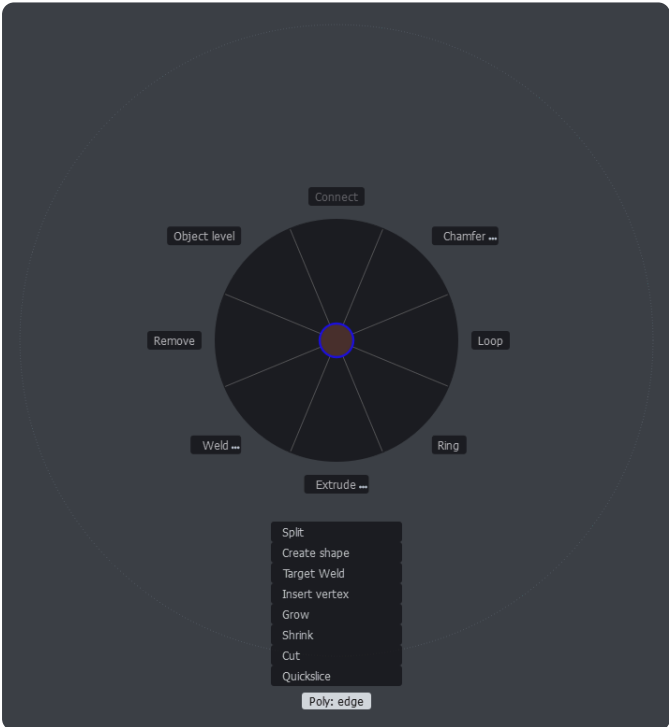
A dial can change its contents with the situation. The built-in **Modelling** dial (Alt+1) is the clearest example: *Convert to Editable Poly* and everything you do next live under the same key.



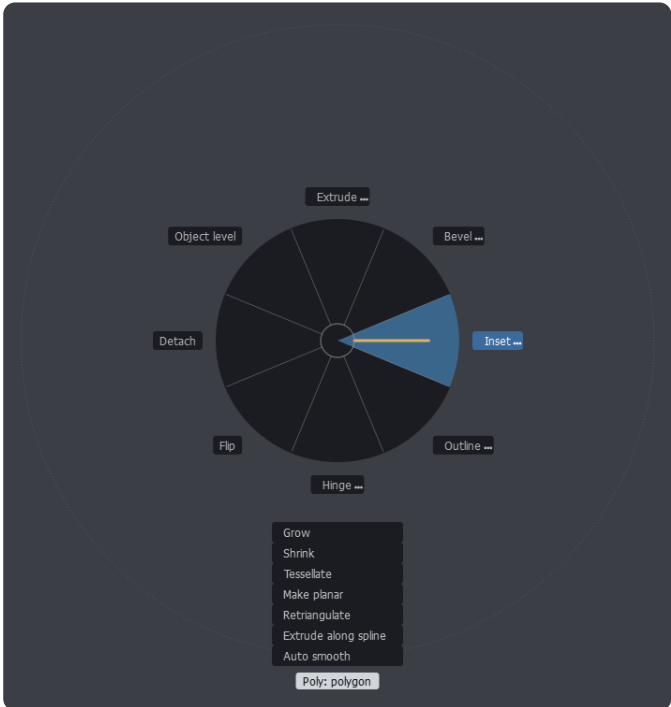
Alt+1 on an Editable Poly at object level: the sub-object levels, Attach, TurboSmooth, Collapse.



The same key at Vertex level: Chamfer, Weld, Connect, Break... NW always returns to object level.



Edge level: Connect, Chamfer, Loop and Ring opposite each other.



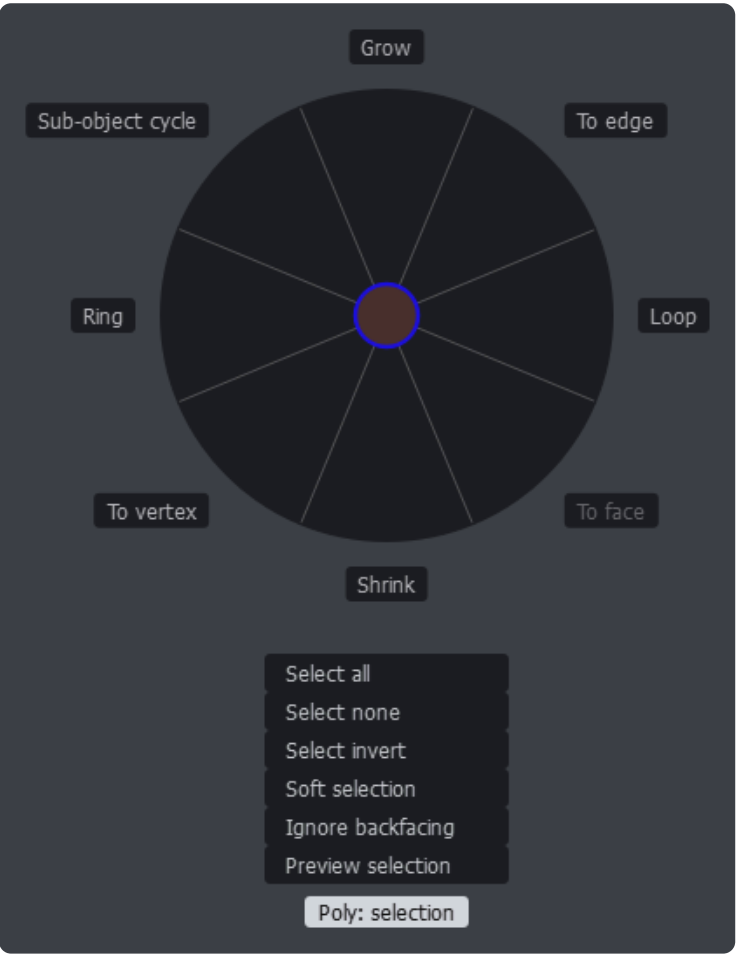
Polygon level, aiming at Inset.

The **same operation keeps the same direction** on an Editable Poly and under an Edit Poly modifier, so your hand does not have to know which one it is working on:

| Level | N | NE | E | SE | S | SW | W | NW |
|-------|---|----|---|----|---|----|---|----|
|-------|---|----|---|----|---|----|---|----|

| Object  | Vertex  | Edge    | Border  | Polygon    | Element         | Attach | + TurboSmooth | Collapse stack |
|---------|---------|---------|---------|------------|-----------------|--------|---------------|----------------|
| Vertex  | Chamfer | Weld    | Connect | Break      | Extrude         | Remove | Target Weld   | Object level   |
| Edge    | Connect | Chamfer | Loop    | Ring       | Extrude         | Weld   | Remove        | Object level   |
| Border  | Cap     | Chamfer | Extrude | Connect    | Create shape    | Shrink | Grow          | Object level   |
| Polygon | Extrude | Bevel   | Inset   | Outline    | Hinge           | Flip   | Detach        | Object level   |
| Element | Detach  | Hide    | Flip    | Unhide all | Hide unselected | Shrink | Grow          | Object level   |

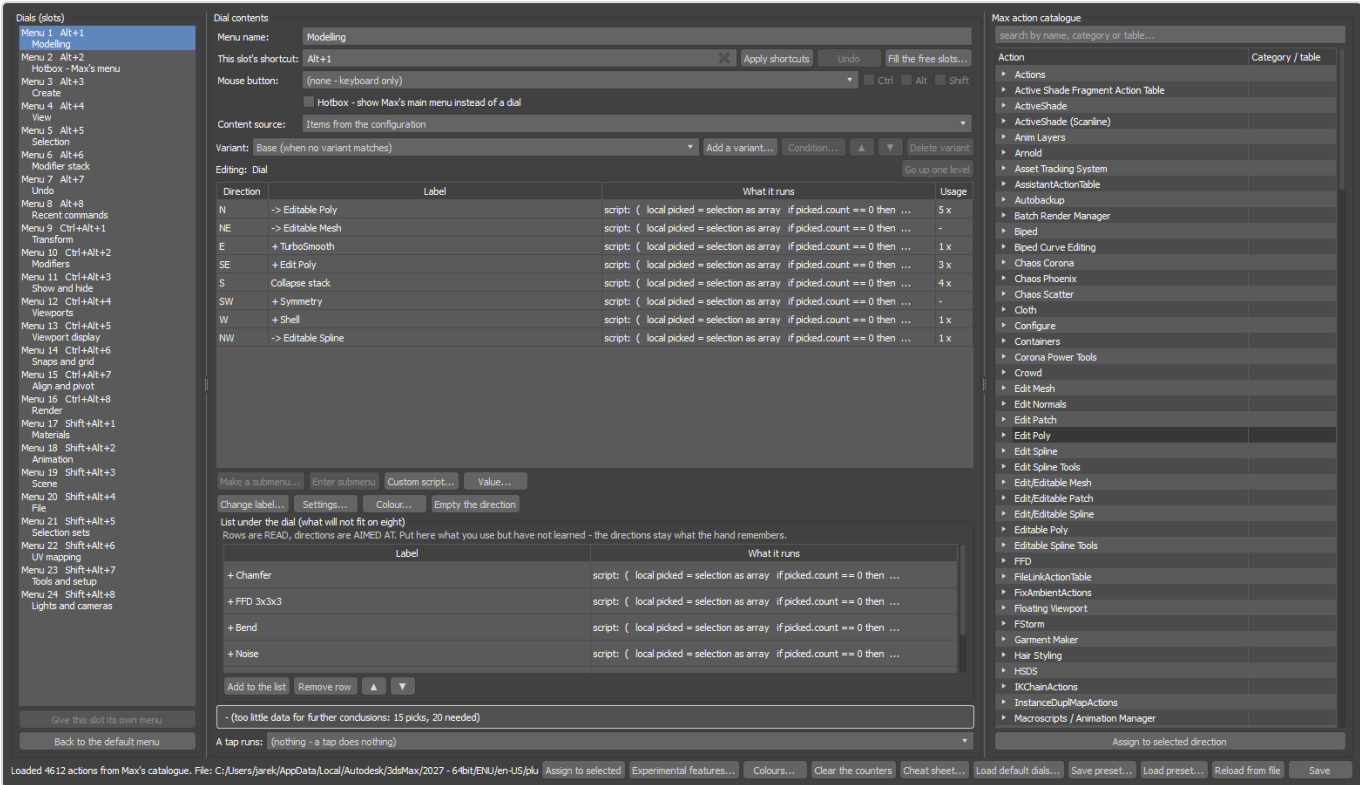
Other dials do the same: Selection offers Grow, Shrink, Loop and Ring at sub-object level; Create turns into shapes when a shape is selected; Lights and cameras turns into camera or light controls; Materials, Animation and Scene change when the cursor is over the Slate editor, Track View or the Scene Explorer. Chapter 11 lists every case.



Alt+5 on an Editable Poly sub-object level: Grow and Shrink opposite each other, Loop and Ring opposite each other.

# 8. The menu editor

**MarkingForge > Menu editor...** Everything about the dials is set here; changes take effect as soon as you press **Save**, without restarting 3ds Max.



The menu editor: dials on the left, the selected dial's directions in the middle, 3ds Max's 4000+ commands on the right.

| Area                      | Use                                                                                                                                                                   |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dials (left)              | The 24 slots with their shortcuts. Pick one to edit it.                                                                                                               |
| Shortcut and mouse button | The key that opens the dial; optionally a mouse button with a modifier (for example Ctrl+Shift+Middle).                                                               |
| Content source            | Items from the configuration, or a live source: modifier stack, undo list, recent commands, selection sets.                                                           |
| Variant                   | The contextual variants of the dial and their conditions (object class, sub-object level, modifier, window under the cursor, selection count, name, layer).           |
| Directions                | N ... NW. Drag a command from the catalogue onto a row, or double-click a command. Custom script, Value (a gesture slider), Submenu and Colour are the buttons below. |
| List under the dial       | Up to twelve extra rows for what does not fit in eight directions.                                                                                                    |
| Catalogue (right)         | Every command 3ds Max knows, including other plugins' - searchable by name.                                                                                           |
| Bottom row                | Cheat sheet (a printable PDF of all dials), Load default dials, Save / Load preset (share your layout), Reload from file.                                             |

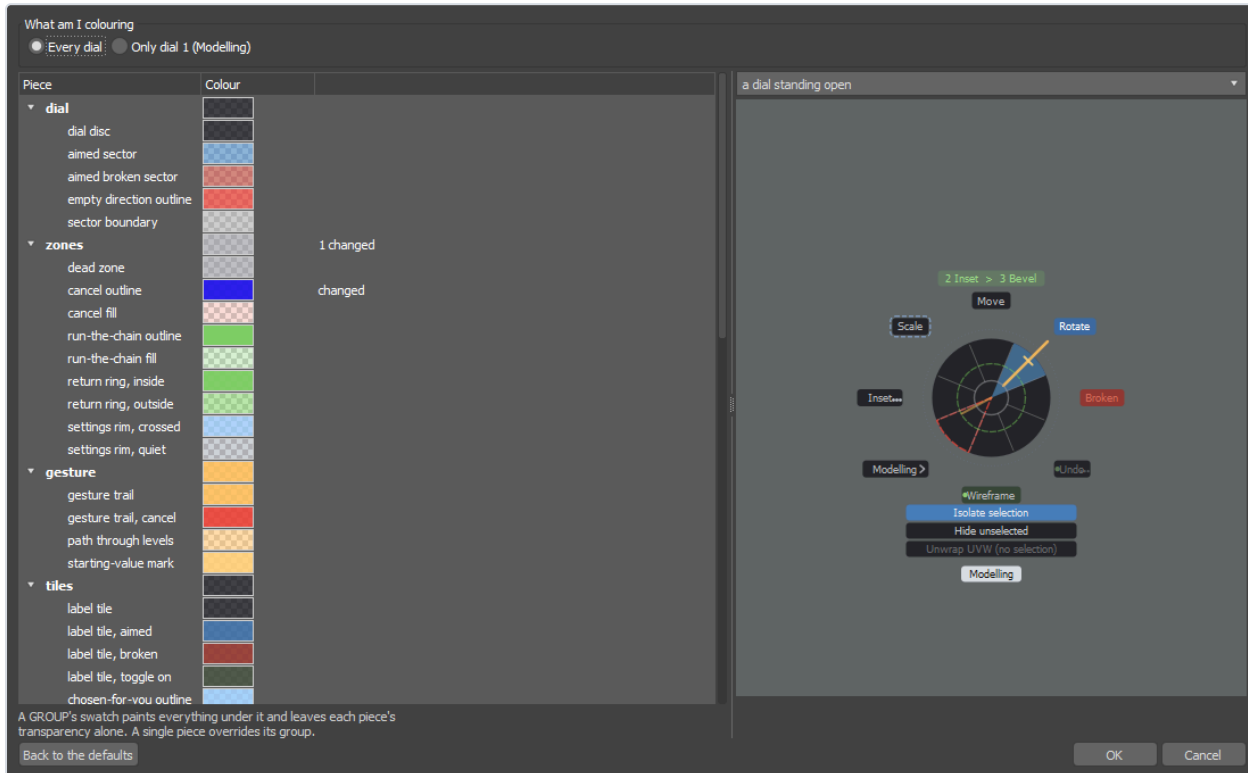
To get the built-in layout back at any time: **Load default dials...** By default it fills only the dials that have no menu of their own.

Closing the editor with unsaved changes (Esc or the window's X) asks whether to **Save**, **Discard** or **Cancel** - work is never thrown away without a question.

## 9. Colours, mouse buttons and experimental features

### Colours

**Colours...** in the editor sets the palette for all dials or for one dial only; **Colour...** under the directions colours a single item. The preview shows the dial at rest, aimed, cancelling and chaining.



The colour editor.

### Opening a dial with a mouse button

Next to the shortcut, pick a mouse button and at least one modifier key. A bare button is refused on purpose: left selects, middle pans and right opens the quad menu - taking one of those away would break 3ds Max.

### Experimental features

Options that change how gestures behave - several picks in one gesture, the menu of the object under the cursor, type-to-search, gesture sliders. Switch them in **Experimental features...** in the editor or in the MarkingForge menu.

These features WORK, but each has a reason for not being on by default. The reason sits under the name - read it first. Switching them all off restores the behaviour from before they existed.

☒ **Direction that sets a value (drag further = more)**

Instead of firing an action, the direction lets you keep dragging, and the distance from the centre sets a number. The release commits it.

WHY EXPERIMENTAL: it needs the current value read from the scene, and that cost has not yet been measured on the path to the first pixel.

☒ **Typing with the menu open searches for actions**

You hold the key, start typing, and the dial shows the best matches from Max's action catalogue.

WHY EXPERIMENTAL: the overlay deliberately does NOT take the keyboard and does not take the focus - working over Slate ME and Track View rests on that. This feature touches that foundation.

☐ **Menus loaded from the scene file**

A menu can travel with a scene - useful when a rigger hands over a scene with a ready-made menu for their rig.

WHY EXPERIMENTAL: somebody else's scene can carry 'script' items, which run with FULL rights. Switch this on only for scenes you trust.

☒ **Multiple picks in one gesture**

Cross the ring and come back to the centre - the item waits in a queue. Releasing the key runs the whole queue as ONE undo.

Releasing at the centre runs the queue; over an EMPTY direction it discards it.

WHY EXPERIMENTAL: it gives meaning to a movement that used to mean nothing. Operation order can matter, and that cannot be checked on the user's behalf - though a wrong order undoes with a single Ctrl+Z.

☒ **Undo dial**

Offers the 'Undo stack' content source: the last eight operations BY NAME. Direction N undoes one, NE two, and so on.

To use it, set Source = Undo stack on some slot and give that slot a shortcut.

WHY EXPERIMENTAL: undoing several steps at once is irreversible other than through Redo, and the operation names are read via MAXScript, because the SDK does not expose the undo stack.

☐ **Menu under the cursor**

The dial describes the object UNDER THE CURSOR rather than the selected one. Picking an item selects it and runs the operation on it - in one undo.

An ALREADY SELECTED object does not change the selection, so a multiple selection stays whole and hovering over any one of its objects acts on the whole group.

WHY EXPERIMENTAL: it is the ONLY one of the three that adds work on the road to the first pixel - the hit test walks the scene. Past 4 ms it switches itself off for the rest of the session and says so in the diagnostics.

☒ **Recent commands dial**

Offers the 'Recent commands' content source: the eight most recently RUN commands, newest first. For REPEATING, as opposed to the undo stack.

Repeating a command moves it to the front, so what you use most stays under your hand by itself.

The history lives in MEMORY and starts afresh when Max starts - it describes what you are doing now, not what you did last week.

WHY EXPERIMENTAL: a record is written after EVERY action that runs. The cost is negligible (one insertion into an eight-element buffer, off the road to the first pixel), but it is a new step in a place that everything passes through.

☒ **Selection sets dial**

Offers the 'Selection sets' content source: the scene's named sets as eight directions. With more than eight it shows the first eight.

WHY EXPERIMENTAL: the sets are read straight from the SDK, but their NAMES are written by the user and end up in the MAXScript code that performs the selection. Quoting is handled (a quote or backslash in a name does not break the code), but the feature is waiting to be confirmed in real work.

OK

Cancel

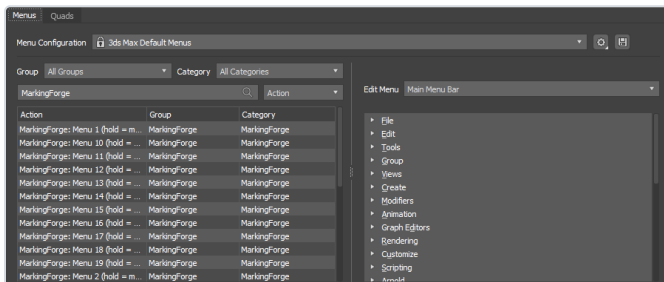
The experimental features, each with the reason it is optional.

# 10. Putting MarkingForge into the 3ds Max interface

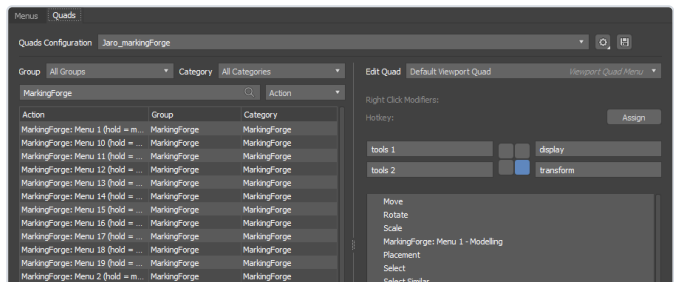
Every MarkingForge command is a standard 3ds Max action in the category **MarkingForge**. You can put any of them on a toolbar, into a menu or a quad menu, or give it a hotkey - exactly like 3ds Max's own commands.

## Into a menu or a quad menu

- 1 Choose **Customize > Menu Editor**.
- 2 On the **Menus** tab (or **Quads** for the right-click quad menu) type **MarkingForge** in the action search field.
- 3 Drag an action from the list into the menu tree on the right.
- 4 Close the editor with **Done**.



Menu Editor - MarkingForge actions ready to drag into a menu.



The same actions on the Quads tab, for the right-click quad menu.

## Onto a toolbar

Choose **Customize > Customize User Interface**, open the **Toolbars** tab, pick the category **MarkingForge** and drag an action onto any toolbar. A button shows the command's name; hovering it shows the description below.

## The commands

| Command                          | What it does                                                                                                                                                        |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Menu editor...                   | Opens a window where you assign actions to dial directions and set keyboard shortcuts.                                                                              |
| Dial cheat sheet (HTML + PDF)... | Collects every dial into one document to read and to print. Along the way it checks which dial would open empty.                                                    |
| Save last chain as an item...    | Takes the last chain you ran (multiple picks) and stores it as ONE direction in a dial of your choice.                                                              |
| Reload configuration             | Reads menus.json again. Useful when the file was edited outside Max.                                                                                                |
| Open menus.json                  | Opens the configuration file in your default text editor. The plugin picks up changes by itself.                                                                    |
| Open configuration folder        | Opens the folder holding menus.json, the shortcut backups and the settings in Explorer.                                                                             |
| Assign keyboard shortcuts...     | Opens 3ds Max's Hotkey Editor. The dials are the actions 'MarkingForge: Menu 1' ... 'Menu 24' - type MarkingForge into its search field.                            |
| Set up standard shortcuts...     | Gives every dial without a shortcut its standard one: Alt+1..8, Ctrl+Alt+1..8, Shift+Alt+1..8. Keys already in use are skipped; your own shortcuts are not touched. |
| Configuration status             | Shows which file is loaded, how many menus it holds and whether it has errors.                                                                                      |
| About MarkingForge               | Shows the plugin's version number and the Qt version it was built against.                                                                                          |

|                               |                                                                                                                                                                                                             |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Documentation                 | Opens the folder with the MarkingForge guides (QuickStart, Reference, Installation and more).                                                                                                               |
| Report a bug...               | Copies your 3ds Max and MarkingForge details to the clipboard and opens an e-mail to <a href="mailto:forgeplugins@gmail.com">forgeplugins@gmail.com</a> .                                                   |
| Suggest a feature...          | Send an idea or a feature request to <a href="mailto:forgeplugins@gmail.com">forgeplugins@gmail.com</a> .                                                                                                   |
| More plugins by the author    | Opens <a href="https://www.artstation.com/loki">https://www.artstation.com/loki</a> in your web browser.                                                                                                    |
| Multiple picks in one gesture | Cross the ring and come back to the centre - the item waits in a queue. The release runs the whole queue as ONE undo.                                                                                       |
| Undo dial                     | Offers the "undo" menu source: the last eight operations by name. Assign it to a slot in the menu editor.                                                                                                   |
| Menu under the cursor         | The dial describes the object UNDER THE CURSOR rather than the selected one. Picking an item selects that object and runs the operation on it - in a single undo.                                           |
| Direction that sets a value   | A "value" direction turns into a slider - the gesture radius sets the number.                                                                                                                               |
| Type-to-search                | Typing while the dial is open turns it into a search over Max's actions.                                                                                                                                    |
| Menus from the scene file     | CAUTION: menus carried in by somebody else's scene. Script items are rejected, but switch this on deliberately all the same.                                                                                |
| Modifier under the cursor     | A dial with the 'Modifier stack' source shows the stack of the object UNDER THE CURSOR rather than the selected one. Requires 'Menu under the cursor' - without it there is nowhere to get the object from. |
| Operation-pair learning       | The plugin remembers what follows what and sets the dial's DEFAULT direction to the most frequent successor. It does NOT move items around - muscle memory rests on that. Session memory only.              |
| Chain as an item              | After you run a chain (multiple picks) the plugin remembers it. You then store it with 'Save last chain...' - with no dialog popping up after every gesture.                                                |
| Recent commands dial          | Offers the 'Recent commands' content source: the last eight you ran, newest first. For REPEATING, as opposed to the undo stack.                                                                             |
| Selection sets dial           | Offers the 'Selection sets' content source: the scene's named sets as eight directions.                                                                                                                     |

The 24 dials themselves are the actions *MarkingForge: Menu 1 ... Menu 24*. They are meant for keys and mouse buttons: a dial lives while the key is held, so a toolbar button cannot hold it open.

# 11. The 24 built-in dials

The layout a fresh installation starts with. Opposite directions hold opposite operations (All / None, Grow / Shrink, Hide / Unhide), so the hand learns one movement and its reverse. The grids show the directions as they sit on screen; the dial's name is in the centre.

Alt+1

Modelling

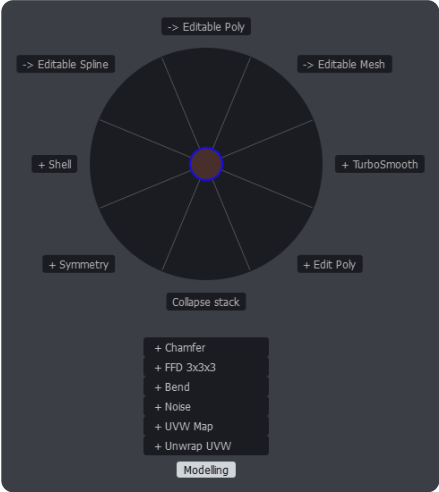
|                          |                       |                        |
|--------------------------|-----------------------|------------------------|
| NW<br>-> Editable Spline | N<br>-> Editable Poly | NE<br>-> Editable Mesh |
| W<br>+ Shell             | Modelling             | E<br>+ TurboSmooth     |
| SW<br>+ Symmetry         | S<br>Collapse stack   | SE<br>+ Edit Poly      |

List under the dial: + Chamfer · + FFD 3x3x3 · + Bend · + Noise · + UVW Map · + Unwrap UVW

Changes with the situation (first match wins):

- **Materials (Slate)** - when cursor over the Slate Material Editor
- **Track View** - when cursor over Track View
- **Particle View** - when cursor over Particle View
- **Scene Explorer** - when cursor over the Scene Explorer
- **Modifier stack** - when cursor over the command panel
- **UV (Unwrap)** - when Unwrap UVW is current in the Modify panel
- **Edit Poly: vertex** - when Edit Poly is current in the Modify panel, sub-object level: Vertex
- **Edit Poly: edge** - when Edit Poly is current in the Modify panel, sub-object level: Edge
- **Edit Poly: border** - when Edit Poly is current in the Modify panel, sub-object level: Border / Segment
- **Edit Poly: polygon** - when Edit Poly is current in the Modify panel, sub-object level: Polygon
- **Edit Poly: element** - when Edit Poly is current in the Modify panel, sub-object level: Element
- **Edit Poly: object** - when Edit Poly is current in the Modify panel
- **Poly: vertex** - when the object is an Editable Poly, sub-object level: Vertex
- **Poly: edge** - when the object is an Editable Poly, sub-object level: Edge
- **Poly: border** - when the object is an Editable Poly, sub-object level: Border / Segment
- **Poly: polygon** - when the object is an Editable Poly, sub-object level: Polygon
- **Poly: element** - when the object is an Editable Poly, sub-object level: Element
- **Poly: object** - when the object is an Editable Poly
- **Spline: vertex** - when a shape (spline) is selected, sub-object level: Vertex
- **Spline: spline** - when a shape (spline) is selected, sub-object level: Spline
- **Spline** - when a shape (spline) is selected
- **Nothing selected** - when nothing is selected

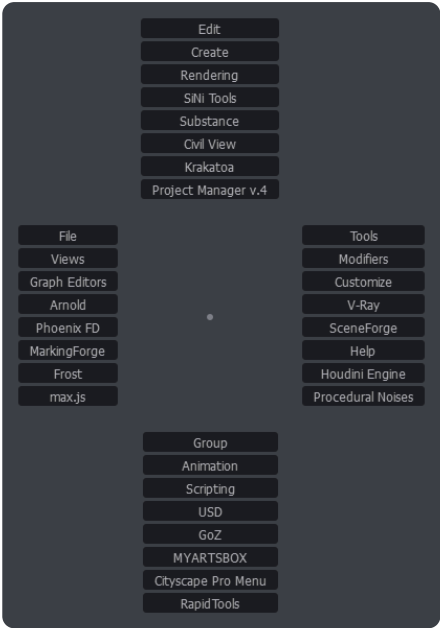
dial 1



**Alt+2 Hotbox - Max's menu**

Shows tiles built from Max's whole main menu - including menus other plugins add. Pick the entry under the cursor.

dial 2



**Alt+3 Create**

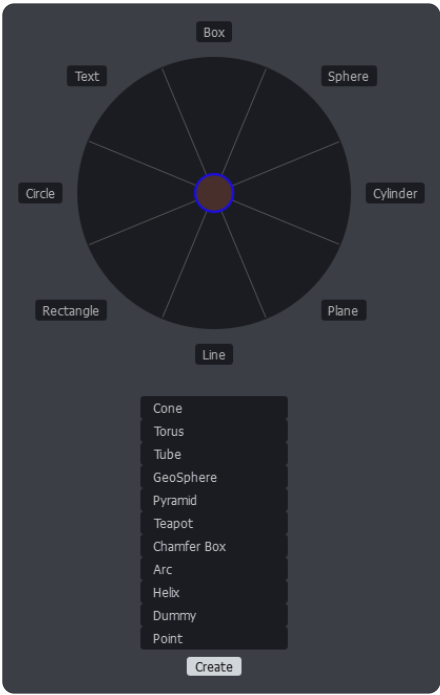
|                 |           |               |
|-----------------|-----------|---------------|
| NW<br>Text      | N<br>Box  | NE<br>Sphere  |
| W<br>Circle     | Create    | E<br>Cylinder |
| SW<br>Rectangle | S<br>Line | SE<br>Plane   |

**List under the dial:** Cone · Torus · Tube · GeoSphere · Pyramid · Teapot · Chamfer Box · Arc · Helix · Dummy · Point

Changes with the situation (first match wins):

- **Create shapes** - when a shape (spline) is selected

dial 3

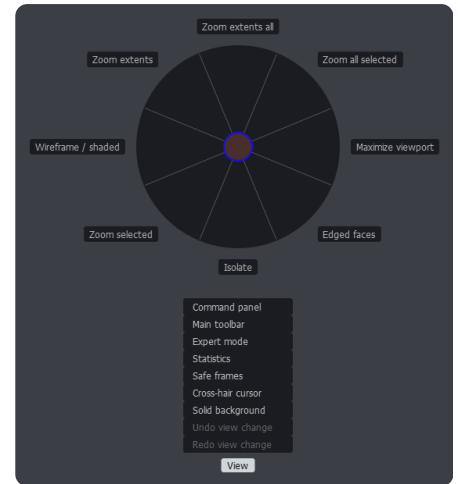


## Alt+4 View

dial 4

|                                |                              |                                |
|--------------------------------|------------------------------|--------------------------------|
| <b>NW</b><br>Zoom extents      | <b>N</b><br>Zoom extents all | <b>NE</b><br>Zoom all selected |
| <b>W</b><br>Wireframe / shaded | <b>View</b>                  | <b>E</b><br>Maximize viewport  |
| <b>SW</b><br>Zoom selected     | <b>S</b><br>Isolate          | <b>SE</b><br>Edged faces       |

**List under the dial:** Command panel · Main toolbar · Expert mode · Statistics · Safe frames · Cross-hair cursor · Solid background · Undo view change · Redo view change



## Alt+5 Selection

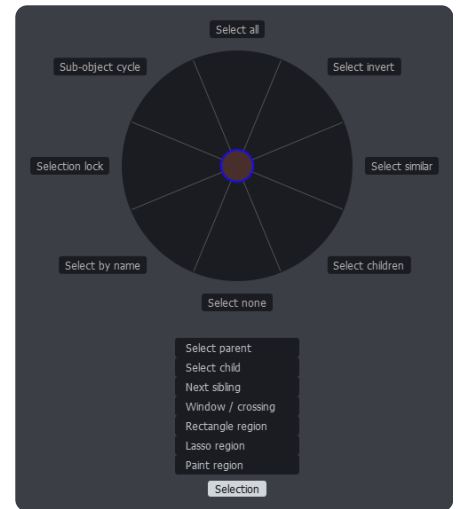
dial 5

|                               |                         |                              |
|-------------------------------|-------------------------|------------------------------|
| <b>NW</b><br>Sub-object cycle | <b>N</b><br>Select all  | <b>NE</b><br>Select invert   |
| <b>W</b><br>Selection lock    | <b>Selection</b>        | <b>E</b><br>Select similar   |
| <b>SW</b><br>Select by name   | <b>S</b><br>Select none | <b>SE</b><br>Select children |

**List under the dial:** Select parent · Select child · Next sibling · Window / crossing · Rectangle region · Lasso region · Paint region

Changes with the situation (first match wins):

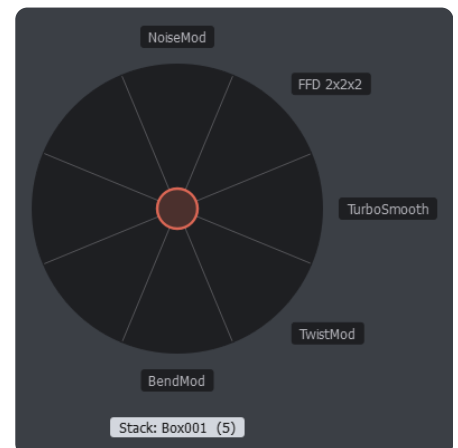
- **Selection** - when Edit Poly is current in the Modify panel, sub-object level: object level
- **Edit Poly: selection** - when Edit Poly is current in the Modify panel
- **Selection** - when the object is an Editable Poly, sub-object level: object level
- **Poly: selection** - when the object is an Editable Poly



## Alt+6 Modifier stack

dial 6

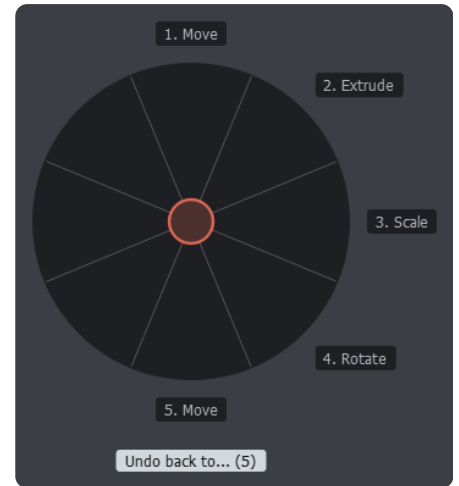
Filled live from the modifier stack of the selected object (top eight).



**Alt+7 Undo**

Filled live from Max's undo list - N undoes one step, NE two, and so on.

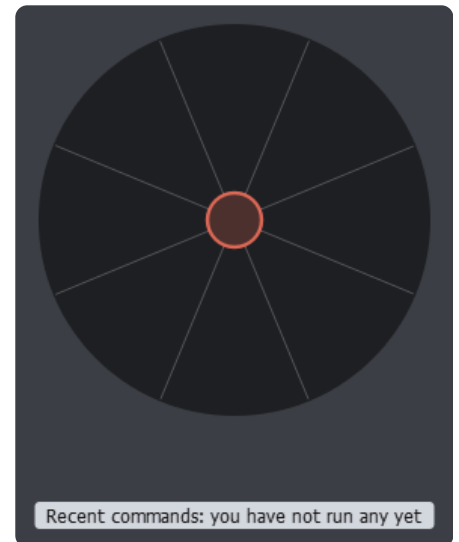
dial 7



**Alt+8 Recent commands**

Filled live from the eight commands you ran most recently.

dial 8

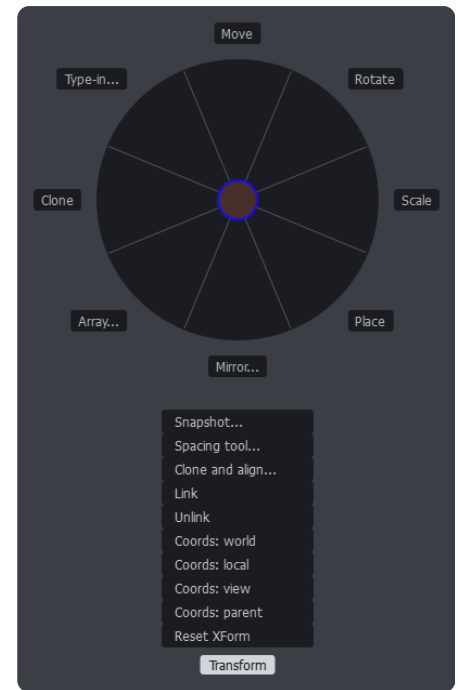


## Ctrl+Alt+1 Transform

|                  |                  |              |
|------------------|------------------|--------------|
| NW<br>Type-in... | N<br>Move        | NE<br>Rotate |
| W<br>Clone       | <b>Transform</b> | E<br>Scale   |
| SW<br>Array...   | S<br>Mirror...   | SE<br>Place  |

**List under the dial:** Snapshot... · Spacing tool... · Clone and align... · Link · Unlink · Coords: world · Coords: local · Coords: view · Coords: parent · Reset XForm

dial 9



## Ctrl+Alt+2 Modifiers

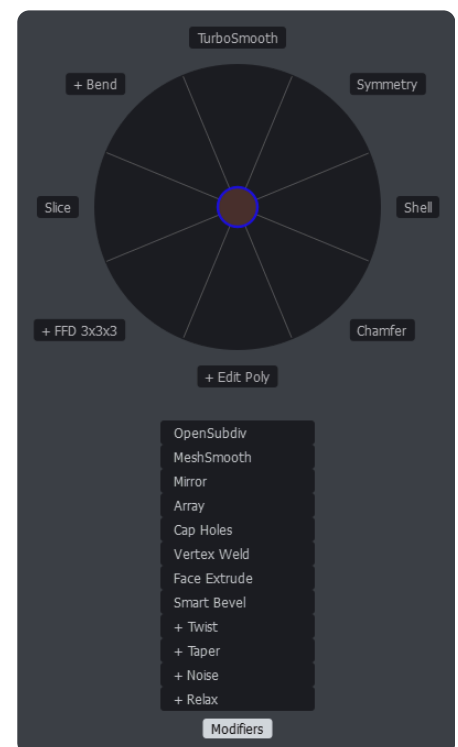
|                   |                  |                |
|-------------------|------------------|----------------|
| NW<br>+ Bend      | N<br>TurboSmooth | NE<br>Symmetry |
| W<br>Slice        | <b>Modifiers</b> | E<br>Shell     |
| SW<br>+ FFD 3x3x3 | S<br>+ Edit Poly | SE<br>Chamfer  |

**List under the dial:** OpenSubdiv · MeshSmooth · Mirror · Array · Cap Holes · Vertex Weld · Face Extrude · Smart Bevel · + Twist · + Taper · + Noise · + Relax

Changes with the situation (first match wins):

- **Shape modifiers** - when a shape (spline) is selected

dial 10



## Ctrl+Alt+3 Show and hide

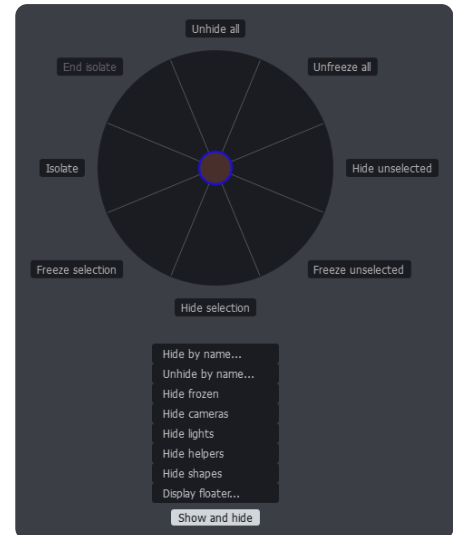
dial 11

|                        |                     |                         |
|------------------------|---------------------|-------------------------|
| NW<br>End isolate      | N<br>Unhide all     | NE<br>Unfreeze all      |
| W<br>Isolate           | Show and hide       | E<br>Hide unselected    |
| SW<br>Freeze selection | S<br>Hide selection | SE<br>Freeze unselected |

**List under the dial:** Hide by name... · Unhide by name... · Hide frozen · Hide cameras · Hide lights · Hide helpers · Hide shapes · Display floater...

Changes with the situation (first match wins):

- **Show and hide** - when the object is an Editable Poly, sub-object level: object level
- **Poly: show and hide** - when the object is an Editable Poly

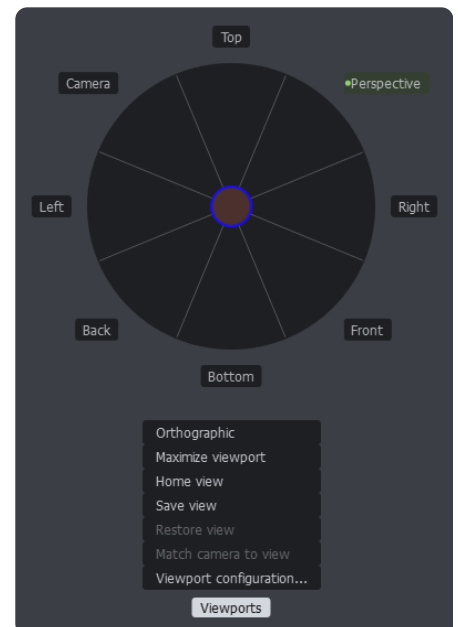


## Ctrl+Alt+4 Viewports

dial 12

|              |             |                   |
|--------------|-------------|-------------------|
| NW<br>Camera | N<br>Top    | NE<br>Perspective |
| W<br>Left    | Viewports   | E<br>Right        |
| SW<br>Back   | S<br>Bottom | SE<br>Front       |

**List under the dial:** Orthographic · Maximize viewport · Home view · Save view · Restore view · Match camera to view · Viewport configuration...



**Ctrl+Alt+5 Viewport display**

dial 13

|                          |                  |                           |
|--------------------------|------------------|---------------------------|
| NW<br>Wireframe / shaded | N<br>Shaded      | NE<br>Wireframe           |
| W<br>See-through         | Viewport display | E<br>Edged faces          |
| SW<br>Edges only         | S<br>Statistics  | SE<br>Facets + highlights |

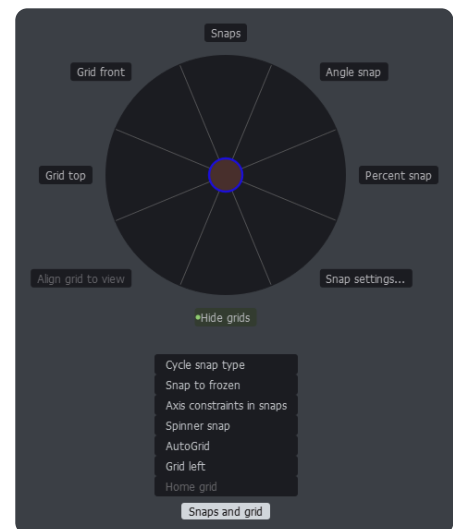
**List under the dial:** Default shading · Realistic with maps · Clay · Bounding box · Lit wireframes · Hidden line · Backface cull · Transparency: best

**Ctrl+Alt+6 Snaps and grid**

dial 14

|                          |                 |                        |
|--------------------------|-----------------|------------------------|
| NW<br>Grid front         | N<br>Snaps      | NE<br>Angle snap       |
| W<br>Grid top            | Snaps and grid  | E<br>Percent snap      |
| SW<br>Align grid to view | S<br>Hide grids | SE<br>Snap settings... |

**List under the dial:** Cycle snap type · Snap to frozen · Axis constraints in snaps · Spinner snap · AutoGrid · Grid left · Home grid

**Ctrl+Alt+7 Align and pivot**

dial 15

|                             |                         |                       |
|-----------------------------|-------------------------|-----------------------|
| NW<br>Working pivot to view | N<br>Quick align        | NE<br>Normal align... |
| W<br>Pin working pivot      | Align and pivot         | E<br>Align to view... |
| SW<br>Place pivot surface   | S<br>Edit working pivot | SE<br>Align camera    |

**List under the dial:** Align... · Place pivot view · Use working pivot · Reset working pivot · Center pivot · Pivot to origin · Reset XForm

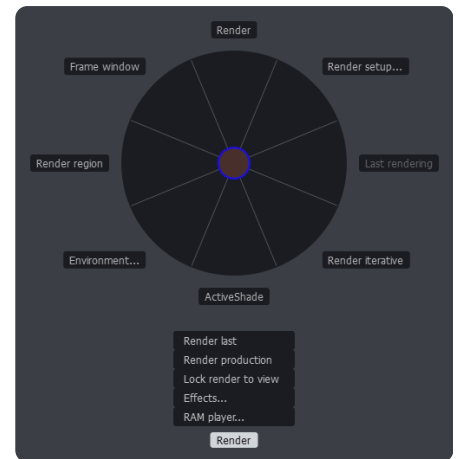


## Ctrl+Alt+8 Render

dial 16

|                             |                         |                               |
|-----------------------------|-------------------------|-------------------------------|
| <b>NW</b><br>Frame window   | <b>N</b><br>Render      | <b>NE</b><br>Render setup...  |
| <b>W</b><br>Render region   | <b>Render</b>           | <b>E</b><br>Last rendering    |
| <b>SW</b><br>Environment... | <b>S</b><br>ActiveShade | <b>SE</b><br>Render iterative |

**List under the dial:** Render last · Render production · Lock render to view · Effects... · RAM player...



## Shift+Alt+1 Materials

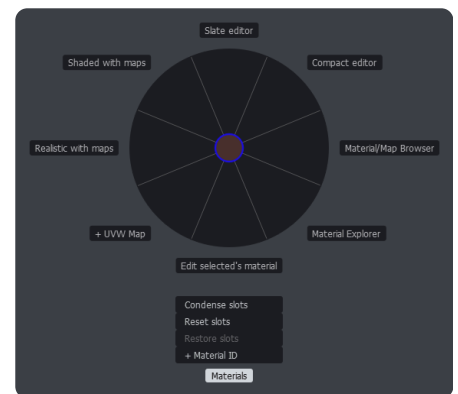
dial 17

|                                 |                                      |                                  |
|---------------------------------|--------------------------------------|----------------------------------|
| <b>NW</b><br>Shaded with maps   | <b>N</b><br>Slate editor             | <b>NE</b><br>Compact editor      |
| <b>W</b><br>Realistic with maps | <b>Materials</b>                     | <b>E</b><br>Material/Map Browser |
| <b>SW</b><br>+ UVW Map          | <b>S</b><br>Edit selected's material | <b>SE</b><br>Material Explorer   |

**List under the dial:** Condense slots · Reset slots · Restore slots · + Material ID

Changes with the situation (first match wins):

- **Materials (Slate)** - when cursor over the Slate Material Editor



## Shift+Alt+2 Animation

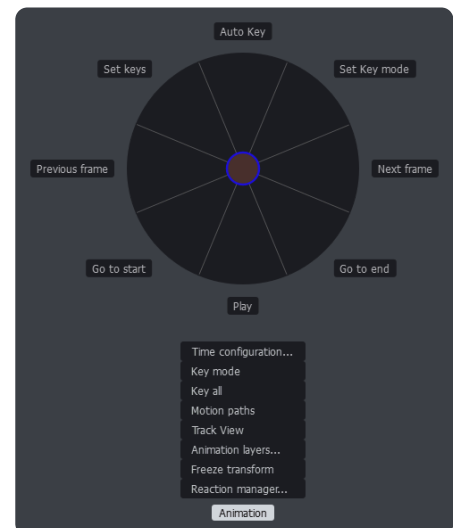
dial 18

|                            |                      |                           |
|----------------------------|----------------------|---------------------------|
| <b>NW</b><br>Set keys      | <b>N</b><br>Auto Key | <b>NE</b><br>Set Key mode |
| <b>W</b><br>Previous frame | <b>Animation</b>     | <b>E</b><br>Next frame    |
| <b>SW</b><br>Go to start   | <b>S</b><br>Play     | <b>SE</b><br>Go to end    |

**List under the dial:** Time configuration... · Key mode · Key all · Motion paths · Track View · Animation layers... · Freeze transform · Reaction manager...

Changes with the situation (first match wins):

- **Track View** - when cursor over Track View



**Shift+Alt+3 Scene**

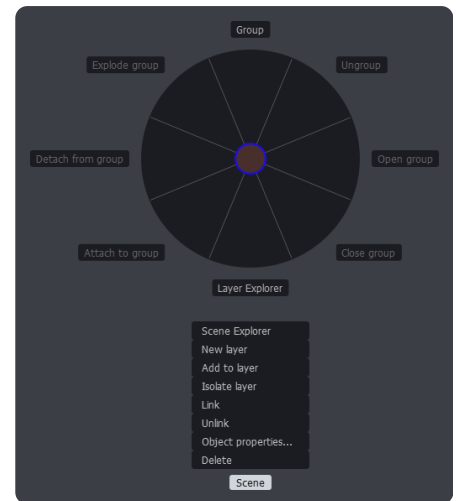
dial 19

|                               |                            |                          |
|-------------------------------|----------------------------|--------------------------|
| <b>NW</b><br>Explode group    | <b>N</b><br>Group          | <b>NE</b><br>Ungroup     |
| <b>W</b><br>Detach from group | <b>Scene</b>               | <b>E</b><br>Open group   |
| <b>SW</b><br>Attach to group  | <b>S</b><br>Layer Explorer | <b>SE</b><br>Close group |

**List under the dial:** Scene Explorer · New layer · Add to layer · Isolate layer · Link · Unlink · Object properties... · Delete

Changes with the situation (first match wins):

- **Scene Explorer** - when cursor over the Scene Explorer

**Shift+Alt+4 File**

dial 20

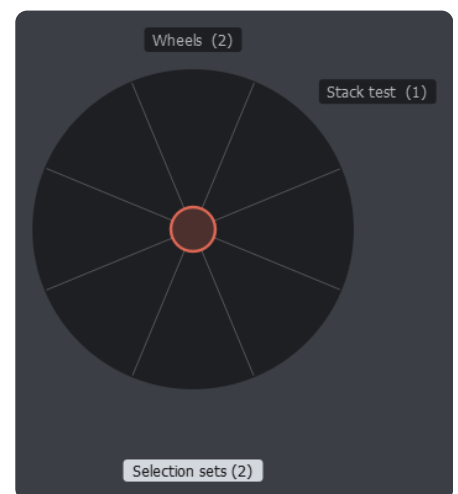
|                        |                     |                              |
|------------------------|---------------------|------------------------------|
| <b>NW</b><br>Hold      | <b>N</b><br>Save    | <b>NE</b><br>Save as...      |
| <b>W</b><br>Merge...   | <b>File</b>         | <b>E</b><br>Save incremental |
| <b>SW</b><br>Import... | <b>S</b><br>Open... | <b>SE</b><br>Save copy as... |

**List under the dial:** Fetch · Export... · Export selected... · Save selected... · Archive... · Summary info... · Reset

**Shift+Alt+5 Selection sets**

dial 21

Filled live from the named selection sets of the scene.



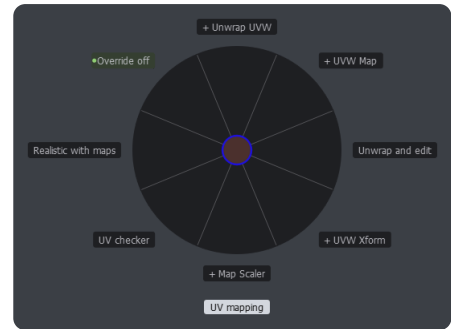
## Shift+Alt+6 UV mapping

dial 22

|                          |                   |                      |
|--------------------------|-------------------|----------------------|
| NW<br>Override off       | N<br>+ Unwrap UVW | NE<br>+ UVW Map      |
| W<br>Realistic with maps | <b>UV mapping</b> | E<br>Unwrap and edit |
| SW<br>UV checker         | S<br>+ Map Scaler | SE<br>+ UVW Xform    |

Changes with the situation (first match wins):

- **UV (Unwrap)** - when Unwrap UVW is current in the Modify panel

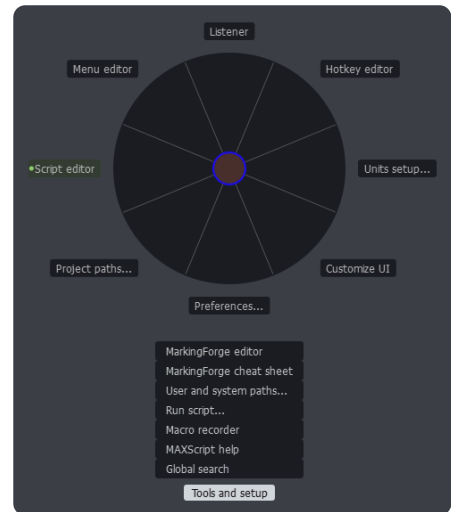


## Shift+Alt+7 Tools and setup

dial 23

|                        |                        |                     |
|------------------------|------------------------|---------------------|
| NW<br>Menu editor      | N<br>Listener          | NE<br>Hotkey editor |
| W<br>Script editor     | <b>Tools and setup</b> | E<br>Units setup... |
| SW<br>Project paths... | S<br>Preferences...    | SE<br>Customize UI  |

**List under the dial:** MarkingForge editor · MarkingForge cheat sheet · User and system paths... · Run script... · Macro recorder · MAXScript help · Global search



## Shift+Alt+8 Lights and cameras

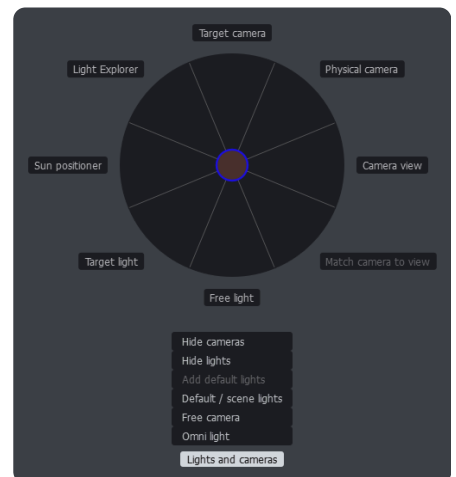
dial 24

|                      |                           |                            |
|----------------------|---------------------------|----------------------------|
| NW<br>Light Explorer | N<br>Target camera        | NE<br>Physical camera      |
| W<br>Sun positioner  | <b>Lights and cameras</b> | E<br>Camera view           |
| SW<br>Target light   | S<br>Free light           | SE<br>Match camera to view |

**List under the dial:** Hide cameras · Hide lights · Add default lights · Default / scene lights · Free camera · Omni light

Changes with the situation (first match wins):

- **Camera** - when a camera is selected
- **Light** - when a light is selected



# 12. Files and settings

| What                 | Where                                                                                 |
|----------------------|---------------------------------------------------------------------------------------|
| The plugin           | %APPDATA%\Autodesk\ApplicationPlugins\MarkingForge                                    |
| Your dials           | %LOCALAPPDATA%\Autodesk\3dsMax\2027 - 64bit\ENU\en-US\plugcfg\MarkingForge\menus.json |
| Shortcut backups     | the same folder, hotkeys\backup_*.hsx - one before every change                       |
| Usage counters, logs | the same folder                                                                       |

**MarkingForge > Open configuration folder** opens it in Explorer; **Open menus.json** opens the file in your text editor. The plugin notices a saved file and reloads it by itself.

To move your setup to another computer, copy menus.json - or use **Save preset...** in the editor, which leaves out paths and usage data.

## 13. Updating and uninstalling

---

### Updating

Install the new version the same way as the first time. The Setup window shows **Update / Repair**; your dials and shortcuts are kept. Restart 3ds Max.

An update cannot leave you without a working plugin: the installer first backs up the installed version (into 3ds Max's temp folder), and if copying the new one fails - a full disk, an antivirus lock - it puts the previous version back and says so. Nothing to do but fix the cause and run it again.

### Uninstalling

- 1 Run the installer (method 1 or 2) and press **Uninstall**, then restart 3ds Max - or close 3ds Max and delete the plugin folder from chapter 12.
- 2 Optional: delete the settings folder `plugcfg\MarkingForge` as well, if you want nothing left behind.

## 14. Troubleshooting

| Symptom                                                  | Cause and fix                                                                                                                                                                                                                                                       |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No MarkingForge menu after the restart                   | Type <code>MarkingForge.version()</code> in the Listener. "Unknown property" = not loaded. Check that the folder is in <code>ApplicationPlugins</code> (chapter 12) and contains <code>PackageContents.xml</code> ; reinstall the Visual C++ 2015-2022 x64 runtime. |
| 3ds Max reports a duplicate plugin at start              | Another copy of <code>MarkingForge.gup</code> exists, usually in the 3ds Max <code>plugins</code> folder. The Setup window lists every copy it finds - delete all but the one in <code>ApplicationPlugins</code> .                                                  |
| A key does not open a dial                               | Check it in <b>Customize &gt; Hotkey Editor</b> : another command may use the same key. Run <b>Set up standard shortcuts...</b> or set the key in the menu editor.                                                                                                  |
| A dial opens empty                                       | It uses a live source with nothing to show (no modifiers, no selection sets). The cheat sheet lists such dials at the top.                                                                                                                                          |
| The dial shows a different menu than expected            | A contextual variant matched. <code>MarkingForge.eventLog()</code> names the variant that won.                                                                                                                                                                      |
| The editor does not open                                 | The error is shown in a message; also check that the <code>tools</code> folder is inside the plugin folder.                                                                                                                                                         |
| A <code>menus.json</code> edited by hand stopped working | <b>MarkingForge &gt; Diagnostics &gt; Configuration status</b> names the line with the error. A broken item disables only that item.                                                                                                                                |

### Listener commands

|                                           |                                                   |
|-------------------------------------------|---------------------------------------------------|
| <code>MarkingForge.version()</code>       | -- is the plugin loaded                           |
| <code>MarkingForge.configSummary()</code> | -- errors and warnings in <code>menus.json</code> |
| <code>MarkingForge.eventLog()</code>      | -- the last gestures: what opened, what ran       |
| <code>MarkingForge.windowInfo()</code>    | -- what the plugin sees under the cursor          |
| <code>MarkingForge.mouseInfo()</code>     | -- mouse-button bindings                          |

MarkingForge sends nothing over the network, does not touch the keyboard outside 3ds Max and changes your shortcuts only when you ask it to.

**Still stuck?** Write to [forgeplugins@gmail.com](mailto:forgeplugins@gmail.com). **MarkingForge > Report a bug...** puts your 3ds Max and MarkingForge details on the clipboard, ready to paste into the e-mail.

## 15. Support, bug reports and feature requests

E-MAIL

[forgeplugins@gmail.com](mailto:forgeplugins@gmail.com)

WEB

<https://www.artstation.com/loki> (ArtStation - more plugins by the author)

One address for everything: questions, problems, **bug reports** and **requests for new features**. Every message is read.

### Reporting a bug

- 1 In 3ds Max choose **MarkingForge > Report a bug...** - it copies your 3ds Max version, MarkingForge version, configuration summary and start-up log to the clipboard. It works even when the plugin did not load.
- 2 Answer **Yes** to open a new e-mail to [forgeplugins@gmail.com](mailto:forgeplugins@gmail.com) - or write from any mail program or web mail.
- 3 Say what you did, what you expected and what happened instead, then paste the details with **Ctrl+V**.
- 4 Add a screenshot if you can. When the problem is in one dial, attach a preset of it (editor > **Save preset...**).

### Suggesting a feature

Choose **MarkingForge > Suggest a feature...** or simply write. Describe what you would like to do and how you do it today - a concrete example from your work says more than a list of wishes. Ideas for new built-in dials, commands and options are all welcome.

Nothing is sent by the plugin itself: the details go to the clipboard and your own e-mail program, so you see everything that leaves your computer. MarkingForge has no network access.